📝 Conditional Statements in Python

Complete Crash Course from Scratch

💡

What are Conditional Statements?

*Conditional statements let your program **make decisions** — just like YOU make decisions every day! They check if something is true or false, and then do different things based on that answer.*

🧠 **Think of it like this:** *"If it's raining, I'll take an umbrella. Otherwise, I'll wear sunglasses." That's a conditional statement in real life!*

📌 1. if Statement

What is it?

The `if` statement is the simplest form of decision-making. It says: **"If this condition is true, do this thing."**

Syntax:

```
if condition:
    # do something
```

⚠️ **Important!** Python uses **indentation** (spaces/tabs) to know what belongs inside the if block. Usually 4 spaces. If you forget indentation, Python will throw an error!

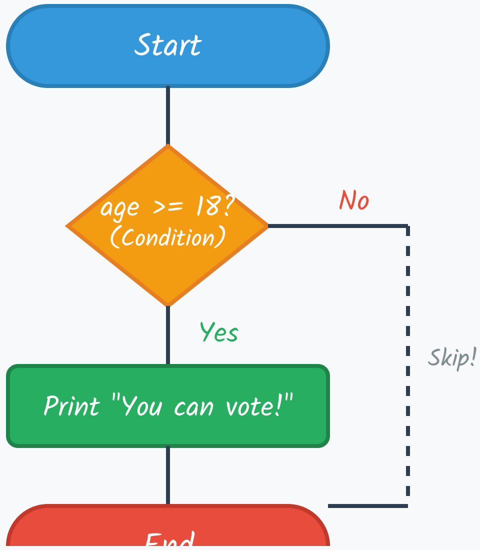
✏️ Code Example

```
age = 18 if age >= 18: print("You can vote! 🗳️")
```

```
Output:
You can vote! 🗳️
```

🌍 Real Life Analogy

🚦 **Traffic Light:** "If the light is GREEN, then GO." You don't do anything if it's red or yellow — you just wait. That's an `if` statement!



Flowchart: if statement

📌 2. if-else Statement

What is it?

The `if-else` statement gives you **two paths**: one if the condition is true, and a different one if it's false. It's like having a Plan A and a Plan B!

Syntax:
```
if condition:
    # Plan A (true)
else:
    # Plan B (false)
```

✏️ Code Example

```
marks = 65 if marks >= 40: print("🎉 You PASSED the exam!") else: print("📚 You need to study more. Keep trying!")
```

```
Output:
🎉 You PASSED the exam!
```

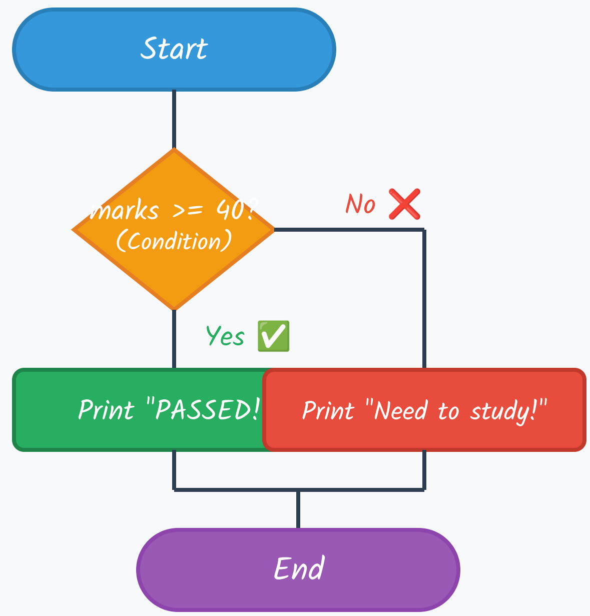
✏️ *Another Example: Even or Odd*

```
number = 7 if number % 2 == 0: print(f"{number} is EVEN 🟢") else: print(f"{number} is ODD 🔴")
```

```
Output:
7 is ODD 🔴
```

🌍 *Real Life Analogy*

🍕 ***Pizza Order:*** *"If I have enough money, I'll buy a large pizza. Else, I'll buy a small pizza." Either way, you get pizza — but the SIZE depends on your condition (money)!*



Flowchart: if-else statement

📌 3. elif (else if) Chains

What is it?

`elif` stands for "**else if**". It lets you check **multiple conditions** one after another. As soon as one condition is true, Python runs that block and skips the rest!

Syntax:

```
if condition1:
    # do A
elif condition2:
    # do B
elif condition3:
```

`# do C`

`else:`

`# do D (if nothing matched)`

💡

🔑 **Key Rule:** Python checks conditions from **TOP to BOTTOM**. The **first true condition wins**! Even if multiple conditions are true, only the first matching block runs.

✏️ Code Example: Grade Calculator

```
score = 85 if score >= 90: print("Grade: A+ 🌟") elif score >= 80: print("Grade: A 👏") elif score >= 70: print("Grade: B 🙂") elif score >= 60: print("Grade: C 😐") elif score >= 40: print("Grade: D 📖") else: print("Grade: F 😢 Study harder!")
```

```
Output:
Grade: A 👏
```

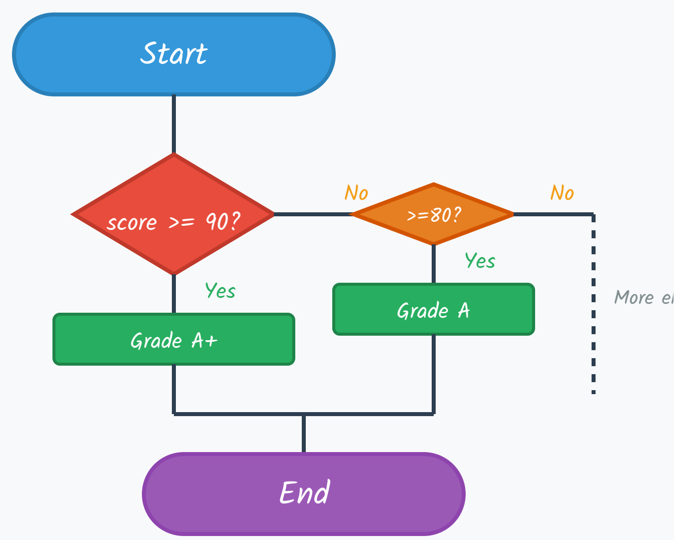
⚠️ **Common Mistake:** Don't write `if` for every condition when you mean `elif`!

❌ `if score >= 90:` ... `if score >= 80:` (checks ALL, may print multiple grades!)

✅ `if score >= 90:` … `elif score >= 80:` *(stops at first match)*

🌍 Real Life Analogy

🌡️ ***Weather Dressing:*** *"If it's above 35°C, wear shorts. Else if it's above 25°C, wear a t-shirt. Else if it's above 15°C, wear a jacket. Else, wear a winter coat." You don't wear ALL of them — you pick the FIRST one that matches!*



Flowchart: elif chain (simplified)

📌 4. Nested Conditions

What is it?

Nested conditions mean putting an `if` statement *inside* another `if` statement. It's like having decisions within decisions — a decision-ception! 🌀

Syntax:

```
if condition1:
    if condition2:
        # do this
    else:
        # do that
else:
    # do something else
```

✏️ Code Example: Movie Ticket Pricing

```
age = 16 is_student = True if age < 18: if is_student: print("🎫 Child Student Ticket: $5") else: print("🎫 Child Ticket: $7") else: if is_student: print("🎫 Adult Student Ticket: $8") else: print("🎫 Adult Ticket: $12")
```

```
Output:
🎫 Child Student Ticket: $5
```

💡 ***Tip:*** *Every time you go one level deeper (nested), you add **4 more spaces** of indentation. Python cares deeply about spacing — it's how it knows which code belongs where!*

✏️ Another Example: Online Shopping

```
total = 120 is_member = True has_coupon = False if
total > 100: if is_member: if has_coupon: print("🎁
Free shipping + 20% off!") else: print("🚚 Free
shipping + 10% off!") else: print("🚚 Free shipping
only!") else: print("💰 Standard shipping: $5")
```

```
Output:
🚚 Free shipping + 10% off!
```

⚠️ ***Don't Nest Too Deep!*** *If you find yourself with 4+ levels of nesting, your code becomes hard to read. Try using* `elif` *or breaking it into functions instead.*

❌ *"If it's Monday, and if it's raining, and if I have an umbrella, and if..." (too confusing!)*

✅ *Use* `and` *operator or flatten with* `elif`*!*

🌍 Real Life Analogy

🏥 ***Hospital Triage:*** *First, check if the patient is conscious. If yes, check if they're bleeding. If yes, check how severe. If severe, send to ER immediately. If not, send to regular ward. Each check is a nested decision based on the previous one!*

📌 5. Ternary Operator (One-Liner if-else)

What is it?

*The **ternary operator** lets you write a simple `if-else` in **just one line**. It's like a shortcut for when you just need to pick between two values!*

Syntax: `value_if_true` `if` `condition` `else` `value_if_false`

✏️ Code Example: Simple vs Ternary

```
# Long way (5 lines) age = 20 if age >= 18: status = "Adult" else: status = "Minor" print(status) # Ternary way (1 line!) 🚀 status = "Adult" if age >= 18 else "Minor" print(status)
```

```
Output (both ways):
```

```
Adult
```

✏️ More Ternary Examples

```
# Find the bigger number a, b = 10, 25 bigger = a if
a > b else b print(f"Bigger number is: {bigger}") #
Check even/odd in one line num = 7 print("Even" if
num % 2 == 0 else "Odd") # Nested ternary (can get
messy!) score = 85 grade = "A" if score >= 90 else
("B" if score >= 80 else "C")
```

```
Output:
Bigger number is: 25
Odd
```

⚠️ **When NOT to use Ternary:** *Don't make it too complex! If your ternary has another ternary inside it (nested ternary), go back to regular if-else for readability. Code is read more often than it's written!*

🌍 Real Life Analogy

☕ **Coffee Order:** *"I'll have coffee **if** it's morning, **else** tea." One sentence, one decision. You don't need a whole conversation — just a quick pick!*

Scenario	Use Regular if-else	Use Ternary
Simple value assignment	✅ Okay	✅ Perfect!
Multiple lines of code	✅ Perfect!	❌ Can't do it
Nested decisions	✅ Perfect!	❌ Gets messy
Readability matters	✅ Better	⚠️ Keep it simple

📌 6. Match-Case Statement

Python 3.10+

What is it?

`match-case` is Python's version of the **switch-case** found in other languages (like C, Java, JavaScript). It's cleaner than long `elif` chains when you're checking the **same variable** against many different values!

Syntax:

```
match variable:
    case value1:
        # do something
    case value2:
        # do something else
```

```
    case _:        # default case (like else)
        # fallback
```

💡 🆕 **New in Python 3.10!** *If your Python version is older, you'll get a* `SyntaxError`*. Check your version with* `python --version`*.*

✏️ Code Example: Day of the Week

```
day = "Wednesday" match day: case "Monday":
print("😴 Start of the week...") case "Tuesday" |
"Wednesday" | "Thursday": print("💼 Mid-week
hustle!") case "Friday": print("🎉 TGIF! Almost
weekend!") case "Saturday" | "Sunday": print("🏖️
Weekend vibes!") case _: print("❓ That's not a
valid day!")
```

```
Output:
💼 Mid-week hustle!
```

✏️ Advanced: Match with Patterns (Tuples)

```
point = (0, 5) # (x, y) coordinates match point:
case (0, 0): print("📍 Origin point!") case (x, 0):
print(f"📏 On X-axis at x={x}") case (0, y):
print(f"📏 On Y-axis at y={y}") case (x, y):
print(f"📍 Point at ({x}, {y})")
```

```
Output:
📏 On Y-axis at y=5
```

💡

🔥 **Cool Features of Match-Case:**

✅ **Multiple values:** `case "A" | "B" | "C":`

✅ **Variable capture:** `case (x, y):` captures values into variables

✅ **Wildcard:** `case _:` catches everything else (like `else`)

✅ **No fall-through:** Unlike C/Java switch, you DON'T need `break`!

🌍 Real Life Analogy

🎮 **Game Controller:** "Match the button pressed: case A → jump, case B → attack, case X → defend, case Y → special move, case _ → do nothing." Each button does exactly ONE thing — clean and organized!

Feature	if-elif-else	match-case
Checking same variable	✅ Works	✅ Cleaner & more readable
Complex conditions (>, <, etc.)	✅ Perfect!	❌ Not designed for this

Pattern matching (tuples, lists)	❌ *Hard to do*	✅ *Built-in superpower!*
Need `break` to prevent fall-through	*N/A*	✅ *No break needed!*
Python version required	*Any version*	*3.10+ only*

📌 7. Quick Reference Cheat Sheet

Statement	Use When...	Syntax
`if`	*One condition, one action*	`if x > 0:` `print("Positive")`
`if-else`	*Two possible outcomes*	`if x > 0: ... else: ...`
`elif` *chain*	*Multiple conditions to check*	`if ... elif ... elif ...` `else:`
Nested `if`	*Decisions depend on other decisions*	`if ...: if ...: ...`
Ternary	*Quick one-line value pick*	`a if condition else b`
`match-case`	*Same variable, many values (3.10+)*	`match x: case "A": ...`

📌 8. Practice Problems

📝 Problem 1: Simple if

Write a program that checks if a number is positive. If yes, print "Positive number!"

Show Solution

📝 Problem 2: if-else

Write a program that checks if a person can drive. Age must be 18+. Print appropriate message.

Show Solution

📝 Problem 3: elif Chain

Create a BMI calculator: Underweight (<18.5), Normal (18.5-24.9), Overweight (25-29.9), Obese (≥30).

Show Solution

📝 Problem 4: Nested Conditions

A theme park ride has rules: Age must be 10+, and if under 15, height must be 120cm+. Write the logic.

Show Solution

📝 Problem 5: Ternary Operator

Use a ternary operator to print "Hot" if temperature > 30, else "Comfortable".

Show Solution

📝 Problem 6: Match-Case (Python 3.10+)

Use match-case to print the number of days in a month (ignore leap years for February).

Show Solution

📝 Challenge Problem: The Smart Door 🚪

A smart door lock has these rules:

- If fingerprint matches AND it's daytime (6am-6pm): Unlock immediately
- If fingerprint matches AND it's nighttime: Ask for PIN code too
- If fingerprint doesn't match: Alarm goes off!

Write this using nested conditions!

Show Solution

📌 9. Summary: Which One to Use?

1 **Start with `if`** — *Do I need to check just ONE thing?*

2 **Add `else`** — *Do I need a "what if not" option?*

3 **Use `elif`** — *Do I have MULTIPLE conditions to check in order?*

4 **Go Nested** — *Does one decision depend on another decision?*

5 **Ternary** — *Is it a super simple one-liner value assignment?*

6 **Match-Case** — *Am I checking the SAME variable against many fixed values?*

🎓 *Made with ❤️ for Learning | Conditional Statements in Python*

Remember: Programming is just teaching computers to make decisions like humans do!